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ESC-113
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An Engineering Analysis of Earthquakes (Sept 2026)

Since before the beginning of humanity, earthquakes have caused devastation on creatures and environments all over the world. What was once thought to be the work of gods was actually massive sections of the earth's mantle, shifting and interacting with each other, causing stress underneath the surface to produce destruction above. These sections are called 'Tectonic Plates', and together they form what we call 'Tectonic Activity'[1]. Some Tectonic Activity is visible to us, like the mountain ranges created by colliding plates, or the volcanoes which spew magma into our cities and ash into our skies. For almost all of history, people have had no way of knowing where or how intense an earthquake would occur at any point in time. That changed with the advancement of modern science, as we now have the tools to measure and analyze tectonic activity in the hopes of predicting and avoiding future disaster.

In this project, a sample of worldwide Earthquakes during a one month span of September, 2026 was selected to be analyzed to find out what real earthquake data can teach us about frequency, magnitude, depth and engineering relevance of seismic events during a specific time period. In order to select the earthquakes which could present a threat to daily life worldwide, only those with a magnitude 2.5 and above were included in the dataset. The main variables which were used in this report were:

[1] "Tectonic Movement - an Overview | Sciencedirect Topics." *Science Direct*, www.sciencedirect.com/topics/engineering/tectonic-movement. Accessed 30 Oct. 2026.

Time: Date and Time in ISO 8601, the international format for representing time[2].

Latitude: The north-south position relative to the equator, measured in degrees (°).

Longitude: The East-West position relative to the earth's meridian, measured in degrees(°).

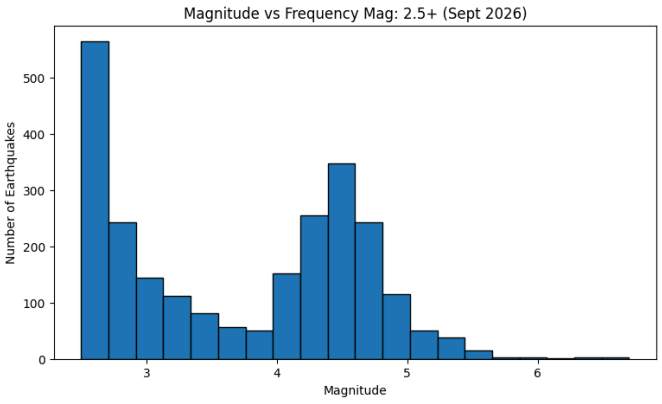
Depth: The distance of the origin of the tectonic activity below the earth's surface, measured in kilometers (km).

Magnitude: A numerical value of the size of the event..

Place: Geographical location.

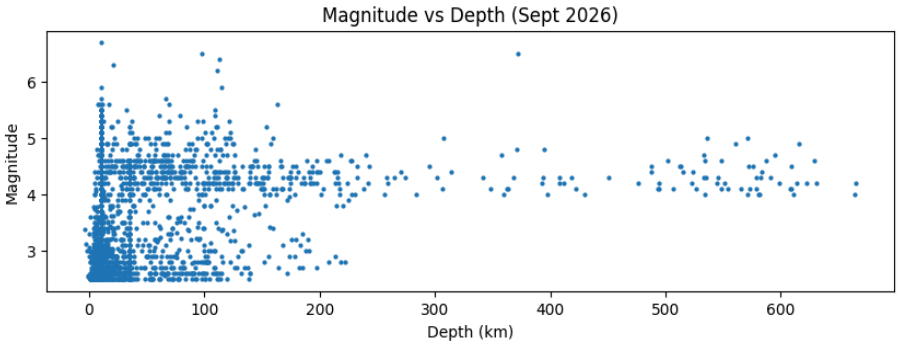
Type: Class of tectonic activity.

Plot Interpretations:



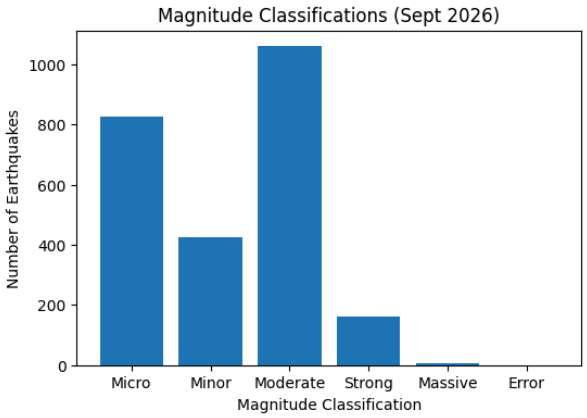
Magnitude vs Frequency: The full range of recorded activity was between 2.5 and 6.7. The graph shows a high occurrence of low magnitude earthquakes around 2.5, with a sharp decline up until around 4.0. Only 166 recorded earthquakes reached/exceeded magnitude of 5.0.

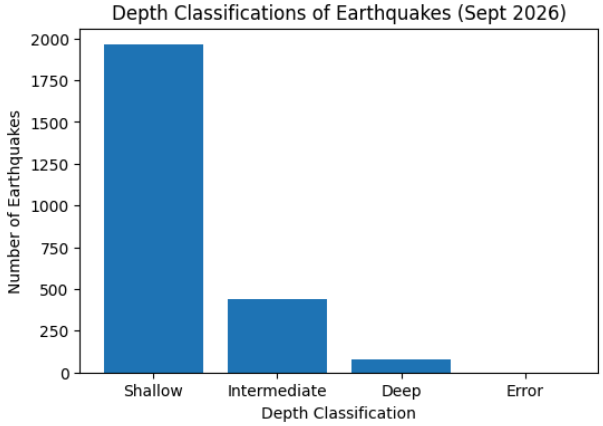
[2] "ISO - ISO 8601 — Date and Time Format." *ISO* , www.iso.org/iso-8601-date-and-time-format.html. Accessed 30 Oct. 2026.



Magnitude vs Depth: Most recorded magnitudes took place on the end of the graph, with the average depth being 53.14km. Only 160 recorded earthquakes reached a depth of over 300km. There appears to be an increase in depth in the 4.0-5.0 magnitude range, however nothing seems to indicate a direct correlation between increasing magnitude and increasing depth. One issue could just be a lack of data on higher magnitude earthquakes, given that they are less common in the dataset. If another month or two was added, it might present a stronger argument.

Magnitude Classifications: The broadest class of earthquake during September was Moderate (Magnitude 4.0-5.5). Relatively very little tectonic activity occurred in stronger magnitude classes, a sharp drop off after Moderate. The Micro class is a close second, and the Minor class after.





Depth Classifications:In the month of September, shallow tectonic activity (<=70km) is the overwhelming majority. While the dataset suggests a large inversely proportionate

relationship between frequency of earthquakes and the depth, the range of time of the dataset is only a month.If the time period of this dataset was extended, a long-term pattern of occurrence could be established, but the one month basis just isn't enough to confirm with certainty.

Engineering Interpretation:

The magnitude range which has the highest concentration of earthquakes in the Magnitude Classification graph is Moderate (4.0 - 5.5), with 1200 earthquakes out of 2477 falling under the class. By using pandas mean value function, the average function value was computed to be 3.71. This would suggest an average range below 4, however by using the pandas mode function, I was able to calculate that the magnitude which appears most in the data set is 4.4, which supports both the Magnitude vs Frequency graph as well as the Magnitude vs Depth graph in terms of visual estimation of the largest magnitude range.

Most Earthquakes are Shallow, with the average depth being 53.14km, which classifies the majority of recorded events in the 'Shallow' depth classification. This is corroborated visually by the Depth Classification graph as well as the Magnitude vs Depth graph.

There does exist a correlation between Magnitude and Depth, but the data really isn't enough to establish any confident answer. In the Magnitude vs Depth graph, as magnitude increases up to the interval of around 4.0-5.0, we do see an increase in depth of these earthquakes. However, as we increase in magnitude further to above 5.0, we see a steady drop in depth, with the only recorded events of this magnitude happening in mostly shallow areas. Only 25 earthquakes measured in magnitude over 5.5, which results in an overall lack of data in higher magnitudes, which leads me to believe that a definitive pattern cannot be established without further data-collection and re-analysis.

The most important engineering datapoint in this set is Data Point #470, the single largest magnitude earthquake in the dataset. It's not only the magnitude which makes this earthquake so important to engineering analysis but mostly its depth. This massive earthquake occurred at 10km depth, 40km above the average depth of the dataset. This pairing of shallow depth and huge magnitude make it extremely dangerous to infrastructure around the area. Luckily, this particular earthquake wasn't near any vital infrastructure, and somehow managed to avoid aftereffects such as tsunami tidal waves. The event was followed by multiple large aftershocks in the region, including two events with 5.1 and 4.8 magnitude respectively[3]. Should this event have happened in an area even slightly closer to vital infrastructure, catastrophic results could have occurred. That's why it's important to analyze and potentially be able to predict events such as #470.

One major limitation of this dataset is just the period of time that it records. Seasonal patterns and tectonic activity cycles can produce data which can be misleading when trying to form patterns on long-term earthquake activity. Having only a month in which to make these predictions is too short of a time to make any definitive answers to broad questions. However, there does exist a frame of reference to work off of in order to build out a system which can answer larger questions and identify broader patterns. Identifying context such as environmental impact and human interference are also necessary to determine the causes and effects of the data.

If I were to improve this into a more complex study, I would choose to broaden the scope of time in which the data was taking place. There just isn't enough context to justify making broad claims. I would also include relevant impacts on nearby infrastructure, perhaps in total

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[3] "M 4.8 - 81 Km ENE of Tadine, New Caledonia." *Earthquake Hazards Program*, earthquake.usgs.gov/earthquakes/eventpage/us6000txuf/executive. Accessed 30 Oct. 2026.

value lost, or maybe in value divided by magnitude to find which earthquakes were damaging particularly because of the place they occurred in.

Bibliography

1. “Tectonic Movement - an Overview | Sciencedirect Topics.” *Science Direct*, www.sciencedirect.com/topics/engineering/tectonic-movement. Accessed 30 Oct. 2026.
2. “ISO - ISO 8601 — Date and Time Format.” *ISO* , www.iso.org/iso-8601-date-and-time-format.html. Accessed 30 Oct. 2026.
3. “M 4.8 - 81 Km ENE of Tadine, New Caledonia.” *Earthquake Hazards Program*, earthquake.usgs.gov/earthquakes/eventpage/us6000txuf/executive. Accessed 30 Oct. 2026.